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Paper 1: Using PBPK to Simulate Target Biopredictive Dissolution Profiles for Long-Acting

Bibliographic Information

- **Title:** Using PBPK to Simulate Target Biopredictive Dissolution Profiles for Long-Acting Injectables - Where to Begin With Critical Bioavailability Attributes?
- **Authors:** Cleary H, Persoons T, Fotaki N, D'Arcy DM
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Quick Take

Pharmacometricians developing long-acting injectables (LAIs) should read this paper to understand how PBPK models can define critical bioavailability attribute (CBA) ranges and generate target dissolution profiles. The study systematically explores particle size, diffusion layer thickness, solubility, diffusion coefficient, and depot volume for methylprednisolone acetate, providing a rational design space for biopredictive in vitro dissolution tests. The multi-step model refinement process, including external validation, offers a replicable workflow for LAI formulation development.

Executive Summary

This study leverages physiologically based pharmacokinetic (PBPK) modeling in GastroPlus to investigate the impact of five CBAs—particle size, solubility, diffusion layer thickness, diffusion coefficient, and depot volume—on the in vivo dissolution and absorption of methylprednisolone acetate LAI. The authors build a PBPK model starting from intravenous disposition analysis, perform a parameter sensitivity analysis, and systematically refine the model to identify plausible CBA value combinations that adequately predict clinical PK data. The final models highlight the dominant roles of effective particle size (agglomeration) and diffusion layer thickness, suggesting that in vivo dissolution occurs in a low-velocity, confined environment with dissolution rates dramatically slower than those generated by current QC methods. Target dissolution profiles span up to 1200 hours versus ~ 90 minutes in compendial tests, providing a critical benchmark for developing biopredictive in vitro methods that support formulation optimization and bioequivalence assessment.

Scientific Context & Motivation

LAIs improve medication adherence but present complex absorption kinetics governed by formulation attributes and local physiological conditions at the injection site. The lack of mechanistic understanding of in vivo dissolution hampers the development of biopredictive in vitro tests, a

gap reflected in the FDA Dissolution Methods Database’s call for method development for many LAI products. This study addresses the need to quantify the influence of CBAs—particularly those not directly measurable in vivo—on systemic exposure, bridging the divide between compendial QC dissolution (rapid, surfactant-based) and prolonged in vivo release. It extends the use of PBPK from traditional oral products to LAI suspensions, incorporating depot dissolution models and inflammation-dependent volume dynamics, thus advancing Model-Informed Formulation Development for complex parenterals.

✂ Methodological Snapshot

The study employs a compartmental IV disposition analysis (PKPlus) followed by a PBPK model of IM depot dissolution and absorption built in GastroPlus. After establishing a baseline simulation, a parameter sensitivity analysis on four CBAs is conducted. Model refinement systematically varies five CBAs at two levels each, with combinations evaluated against observed clinical PK for methylprednisolone. Best models are selected based on visual and statistical fit, and their simulated in vivo dissolution profiles are extracted as target design spaces. One model undergoes external validation with multi-dose RA patient data.

☒ Structural Model Breakdown

The model comprises a two-compartment systemic disposition structure (central and peripheral) linked to a depot dissolution sub-model. The IV disposition model, using methylprednisolone sodium succinate data, established the distribution and elimination parameters (CL , V_c , V_p , Q). For the IM LAI, the depot sub-model assumes that the injected suspension contains spherical particles of defined size distribution; dissolution follows a diffusion-layer-controlled process where the dissolution rate depends on particle surface area, solubility, diffusion coefficient, and diffusion layer thickness. The dissolved drug then undergoes absorption into the systemic circulation through the muscle tissue, governed by tissue permeability and blood flow. Conversion of the acetate prodrug to the active methylprednisolone is implicitly handled by testing inputs for both species, rather than through an explicit depot metabolism compartment. Depot volume can be static or modulated by an inflammation time profile, though here a reduced constant volume (1 mL) was used. Four final model variants (A–D) differed in the specific combination of solubility, particle size (MPR vs PSD), and DLT (500 μm vs 10,000 μm), while retaining the same systemic PK parameters.

Detailed Methodological Analysis

Modeling Approach

A PBPK model for IM injection was developed in GastroPlus v8.3.3 using the built-in depot dissolution and absorption model. IV disposition parameters (2-compartment) were estimated with PKPlus from methylprednisolone sodium succinate data. The IM model incorporates drug physicochemical properties, formulation attributes, and physiological parameters. Dissolution within the depot is simulated using a particle-size-dependent model with a static diffusion layer.

Absorption from depot to systemic circulation is driven by the regional permeability, blood flow, and drug properties. A separate inflammation-dependent depot volume profile was tested but ultimately not included.

Data Sources

IV PK data (dataset A) from literature used for disposition analysis. IM PK data (dataset B, single dose) from an FDA review served as the primary training/validation target. An independent dataset C (multiple 120 mg doses in RA patients) was used for external validation (citation [30]). Drug property values were sourced from regulatory documents, product monographs, and in silico predictions (ADMET Predictor). Particle size distribution was extracted from microscopy data in literature (Benzon et al.).

Estimation Methods

PKPlus was used to fit 1-, 2-, and 3-compartment models to IV data; model selection was based on AIC, SIC, and residual analysis. PBPK simulations were deterministic; model parameter values were either literature-based, predicted by ADMET Predictor, or explored across two pre-specified levels. No population parameter estimation or maximum likelihood methods were used for the PBPK step.

Model Evaluation

Predictive performance was assessed by fold error (FE), average fold error (AFE), and absolute average fold error (AAFE) for C_{max} , T_{max} , $AUC_{0-\infty}$, AUC_{0-t} . A model was considered acceptable if key PK parameters fell within 2-fold of observed. Visual inspection overlaid simulations with observed mean data. External validation applied the same metrics to an independent dataset.

Covariate Analysis

No formal covariate analysis was performed. Body weight was fixed at 70 kg, and no variability was modeled. The impact of physiological factors (inflammation, depot volume reduction) was explored as deterministic scenarios rather than statistical covariates.

Statistical Rigor Assessment

The IV disposition model selection appropriately used information criteria (AIC, SIC) and residual plots, favoring a parsimonious 2-compartment model. For the PBPK models, the evaluation metrics (FE, AFE, AAFE) are standard and relevant for assessing model adequacy. However, the model refinement was based on a deterministic manual grid search over two-level factors, which, while systematic, does not provide formal parameter identifiability analysis or confidence intervals for CBA estimates. Only mean data was used for fitting in datasets A and B (no individual-level variability), precluding true population PK analysis and assessment of between-subject variability. The external validation on dataset C, which includes SD error bars, provides visual evidence of predictive ability at the population mean level, but a formal VPC or NPDE was not performed. The non-inclusion of an inflammation factor was justified by its minimal impact, which simplifies the model and is reasonable given the data resolution.

☒ Key Findings

Parameter sensitivity analysis and iterative model refinement revealed that particle size (mean particle radius or particle size distribution) and diffusion layer thickness are the most influential CBAs governing C_{max} , T_{max} , and overall profile shape. Best-performing models (A–D) accurately predicted observed data when using either a large *MPR* (500 μm) with a 500 μm *DLT* or the undiluted *PSD* with a 1 cm *DLT*, indicating that effective in vivo particle size is much larger than measured ex vivo due to agglomeration. Solubility markedly affected C_{max} , while diffusion coefficient and depot volume had minimal impact. Importantly, simulated in vivo dissolution profiles showed 80–100% drug released only after 1200 hours, in stark contrast to in vitro QC methods that dissolve 90% in 90 minutes, underscoring the need for low-velocity, spatially constrained dissolution conditions for biorelevance. External validation with rheumatoid arthritis patient data confirmed model C’s predictive ability.

🔗 Clinical & Regulatory Implications

The identified CBA ranges and target dissolution profiles provide a scientific benchmark for developing biorelevant in vitro dissolution tests for methylprednisolone acetate LAI. Regulatory scientists can use these target profiles to evaluate whether proposed QC methods reflect in vivo release kinetics, potentially supporting biowaivers for minor formulation changes. The findings emphasize that effective particle size (agglomeration) in the depot environment is much larger than the primary particle size, informing formulators about the impact of particle engineering and vehicle composition. The external validation in RA patients suggests the model can support clinical trial simulations and dosing regimen design for this LAI. More broadly, the workflow can be adapted to other LAI products, contributing to Model-Informed Drug Development strategies for complex generics under GDUFA.

Strengths & Limitations

Strengths

- Clear, EMA guideline-informed workflow that integrates IV disposition analysis, baseline PBPK, PSA, and systematic model refinement, ensuring a robust model development process.
- Systematic exploration of five CBAs using plausible literature- and prediction-derived ranges, revealing important parameter interactions (e.g., *DLT* and particle size coupling).
- Generation of target in vivo dissolution profiles that provide a concrete design space for biopredictive in vitro test development, directly linking to regulatory quality needs.
- External validation using an independent clinical dataset (multiple-dose, RA patients) increases confidence in the model’s predictive capability beyond the training data.
- Critical assessment of in vitro–in vivo disconnect in dissolution timescales, motivating development of more discriminative, biorelevant dissolution methods for LAIs.

Limitations (Acknowledged by Authors)

- True in vivo solubility may lie outside the estimated range; local IM fluid composition is not fully characterized.

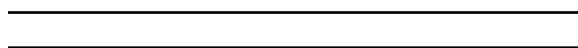
- Datasets A and B contained only mean PK data without variability measures, limiting assessment of inter-individual variability capture.
- The values of some CBAs (e.g., effective *DLT*, in vivo *MPR*) remain uncertain, and multiple parameter combinations yielded adequate fits, indicating an element of non-identifiability.

Limitations (Expert Review)

- The model does not incorporate a mechanistic description of butyrylcholinesterase-mediated conversion of methylprednisolone acetate to methylprednisolone within the depot versus systemic circulation, which could influence local solubility and dissolution.
- Inflammation factor implementation was simplistic and ultimately omitted; a more dynamic, drug- and formulation-specific inflammatory response model might improve depot volume predictions.
- The PBPK platform's default depot dissolution model assumes a spherical particle geometry with homogeneous diffusion layer, which may not fully capture complex depot morphology or drug aggregation.
- The study used a limited number of in vivo datasets (2 for model building/validation), and the external validation dataset, though independent, still reflects a specific patient population (RA).

Generalizability

The workflow is broadly applicable to other poorly soluble LAI suspensions, as it demonstrates how to derive target dissolution profiles from PK data when formulation-specific in vivo parameters are unknown. Caution is needed when extrapolating specific CBA values (e.g., depot volume reduction factor) to other drugs or injection sites without additional mechanistic justification. The approach is most reliable for low-risk, exploratory purposes guiding in vitro method development, consistent with PBBM risk-based frameworks.



Figures & Tables

- **Figure 1:** Schematic illustrating critical formulation and physiological attributes (CBAs) affecting LAI bioavailability, including depot formation, particle dissolution, and absorption.
 - *Significance:* Provides conceptual framework linking the five modeled CBAs to the in vivo drug release process, essential for understanding model parameterization.
- **Figure 2:** Workflow diagram following EMA PBPK guidelines, showing the stepwise approach from IV disposition analysis to external validation.
 - *Significance:* Outlines the modeling strategy and experimental tiers, demonstrating adherence to regulatory model development standards.
- **Figure 3:** Series of simulated plasma concentration–time profiles from 19 model variants (altering solubility, particle size, *DLT*, *DV*) overlaid on observed data.
 - *Significance:* Shows the parameter space exploration and identifies models A–D that adequately capture the observed PK, highlighting the interplay between CBA values.
- **Figure 4:** Target in vivo dissolution profiles (% dissolved over 1200 h) for models A–D, derived from the simulated drug release curves.

- *Significance*: Defines the biorelevant dissolution design space; contrasts with current rapid in vitro QC methods and sets target ranges for test development.
 - **Figure 5**: External validation plot: simulated (model C) vs. observed methylprednisolone concentrations from a multi-dose study in RA patients.
 - *Significance*: Confirms model C's predictive accuracy in an independent clinical population, supporting its use for clinically relevant dissolution profile generation.
 - **Table 1**: Summary of drug physicochemical and pharmacokinetic input parameters (e.g., $\log P$, solubility, CL , V_d) with sources.
 - *Significance*: Documents the initial model parameterization and data provenance, critical for reproducibility and understanding baseline assumptions.
 - **Table 2**: High- and low-level values explored for five CBAs (solubility, diffusion coefficient, MPR/PSD , DLT , depot volume) during model refinement.
 - *Significance*: Defines the examined parameter space and the rationale for each extreme value, anchoring the systematic sensitivity and model selection steps.
 - **Table S4**: Contains model evaluation metrics (FE, AFE, AAFE) for all simulations, including the final four models A–D and the external validation.
 - *Significance*: Supports the model selection process and demonstrates that the chosen models meet predefined predictive performance criteria.
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Future Directions

Characterize the composition (protein content, viscosity) and volume of human IM interstitial fluid to refine solubility and diffusion parameters. Develop more sophisticated depot models that account for particle agglomeration dynamics, non-spherical geometries, and local enzymatic conversion. Extend the workflow to other LAI products with different release mechanisms (e.g., in situ gels, microspheres) to validate the approach. Integrate population PBPK with clinical variability to assess bioequivalence risk and set clinically relevant dissolution specifications.

Expert Commentary

This paper addresses a pressing industrial need: translating the complex in vivo behavior of LAIs into a rational in vitro test design space. The finding that effective particle size is dramatically larger in vivo than ex vivo PSD measurements is a crucial insight—formulators and pharmacometricians should not naively use raw PSD data for LAI PBPK without considering agglomeration. The stark contrast between 1200-hour in vivo dissolution and 90-minute QC dissolution highlights a fundamental disconnect; the field must move toward low-flow, small-volume, non-sink dissolution methods for LAIs. From a model credibility perspective, the use of multiple plausible parameter sets (models A–D) to define a design space is clever and acknowledges model non-identifiability, though it underscores the need for more definitive characterization of key CBAs like effective solubility in the IM milieu. The simple handling of the prodrug conversion is a notable simplification—future models should explicitly mechanistically describe local depot metabolism to enhance physiological relevance. Overall, this is a well-structured, industrially relevant application of PBPK that sets the stage for more biopredictive LAI dissolution testing.

Bottom Line

PBPK modeling of LAI suspensions can effectively map critical bioavailability attributes to systemic exposure and generate target dissolution profiles that illuminate the vast differences between current QC methods and in vivo release. This work gives formulators a practical framework to systematically identify in vitro testing conditions that better reflect the low-velocity, agglomeration-prone depot environment, ultimately supporting more biopredictive dissolution methods and accelerating LAI development.

☒ Figures

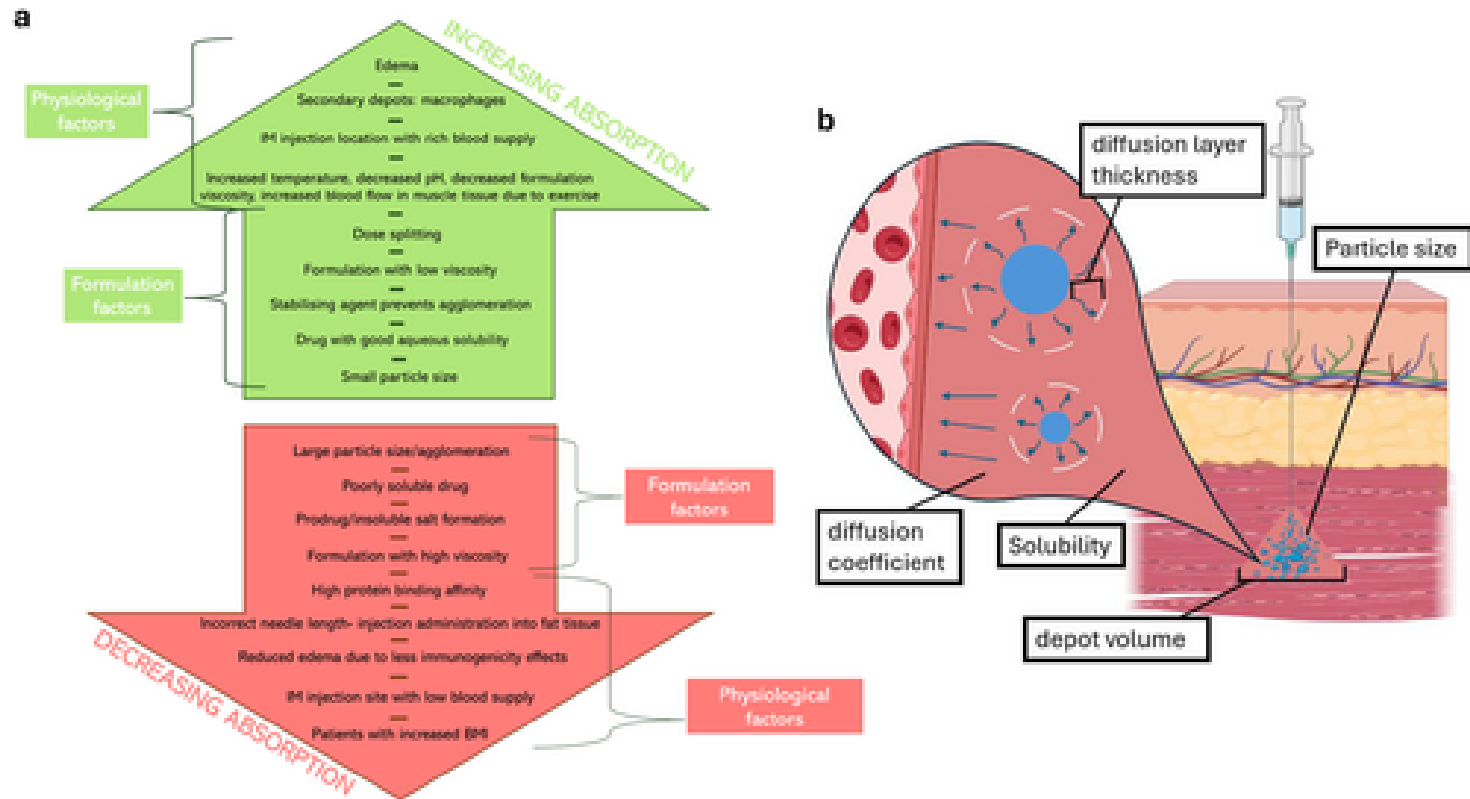


Figure 1: (a) A depiction of critical formulation and physiological attributes relating to long-acting injectable preparation bioavailability.
(b) The five critical bioavailability factors

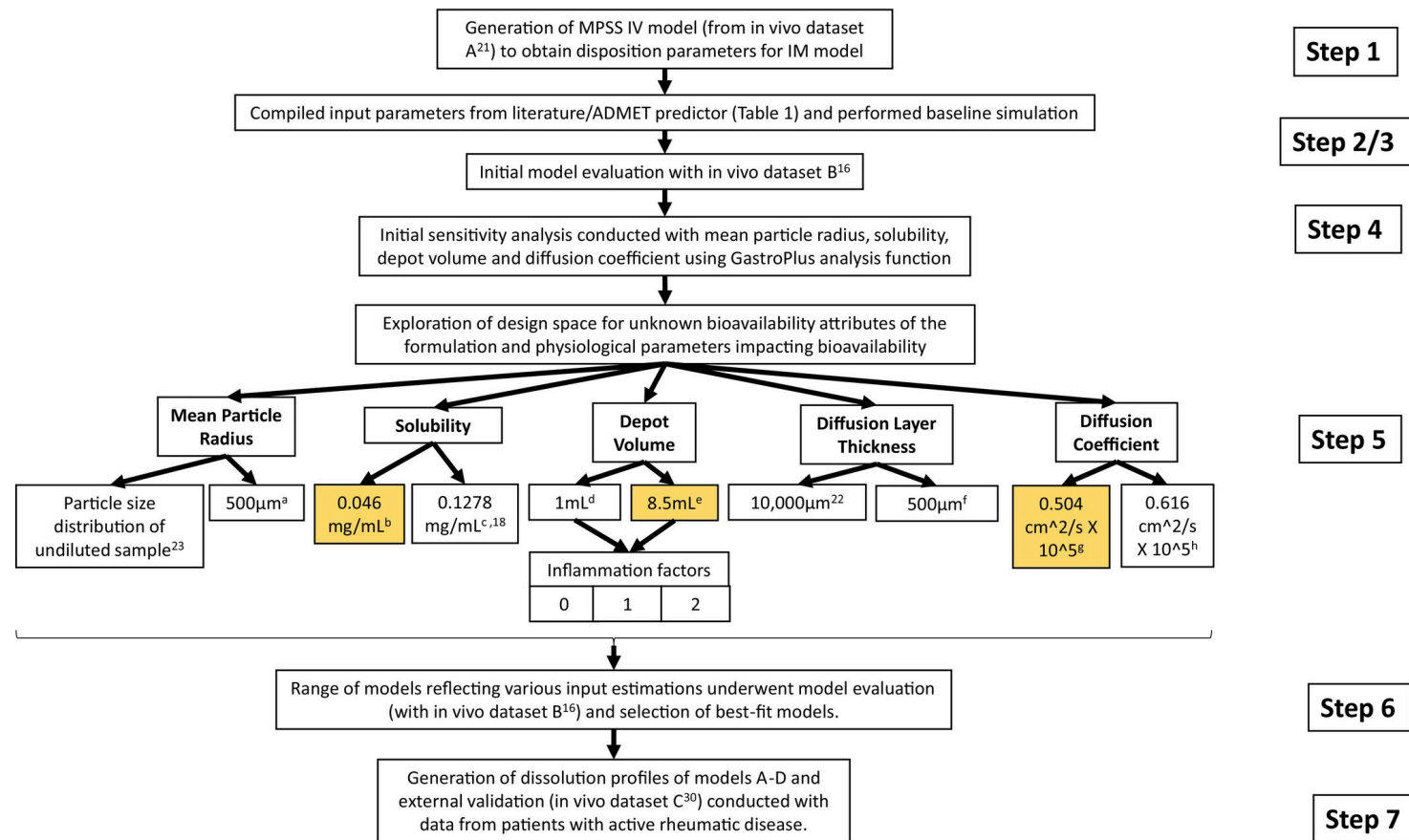


Figure 2: The workflow approach used for the modeling and simulation studies, guided by the EMA's 'Guideline on the reporting of physiologically based pharmacokinetic mode

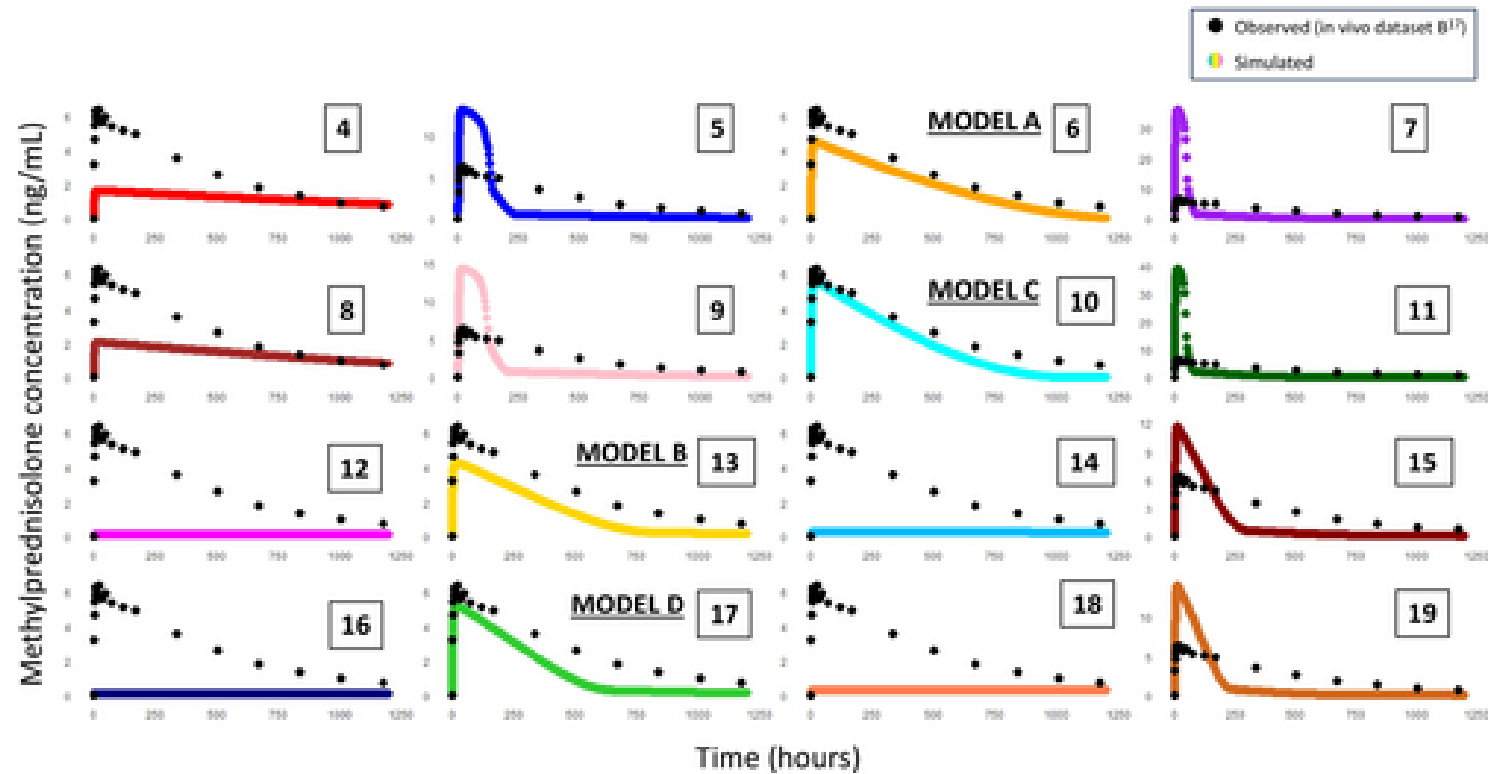


Figure 3: A series of simulations conducted using baseline model inputs from Table 1 and alterations as listed in Table 2, with observed data from in vivo dataset B in black

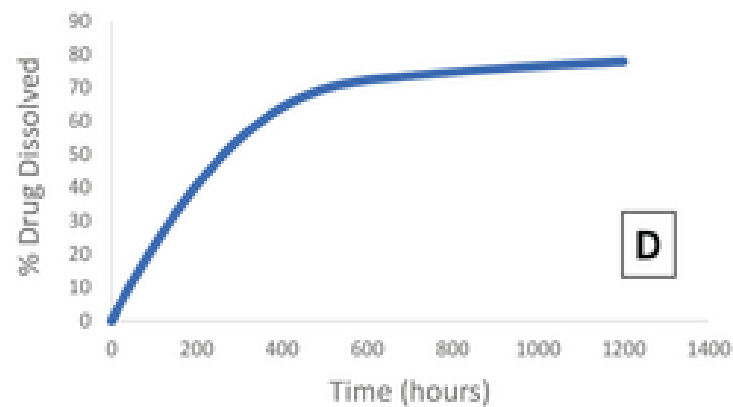
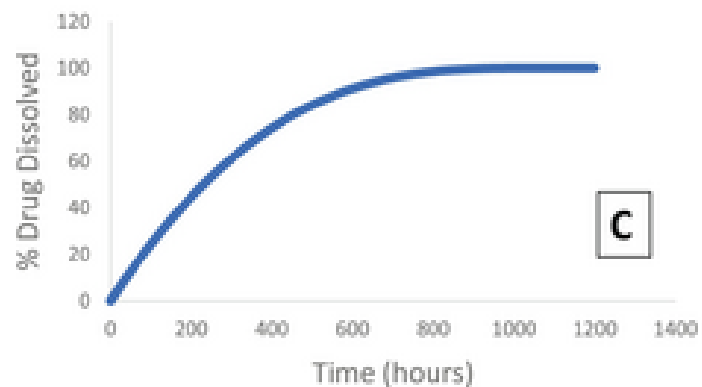
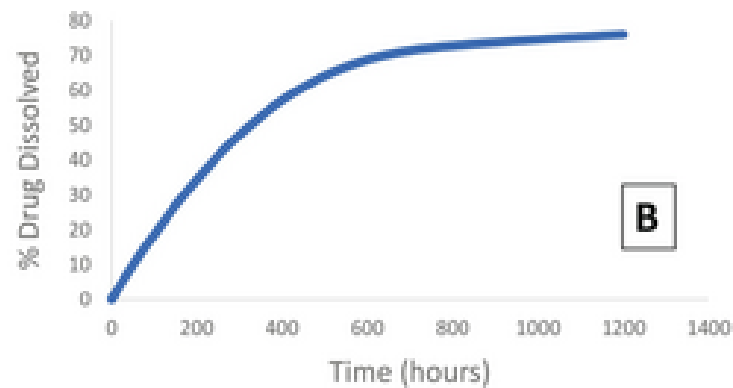
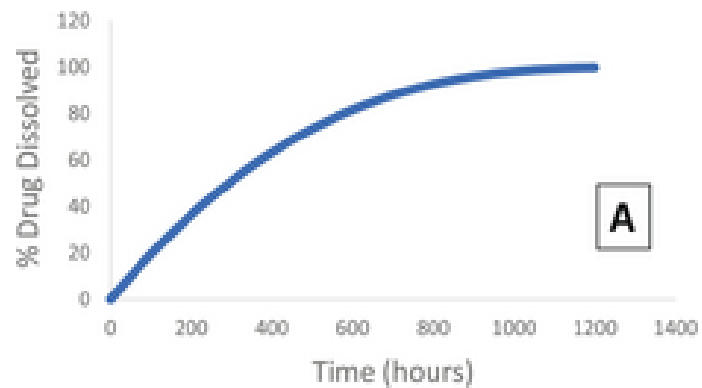


Figure 4: Target dissolution profiles (% drug dissolved versus time) based on simulated profiles. Target profiles from model A and C both reach 100% drug dissolved within

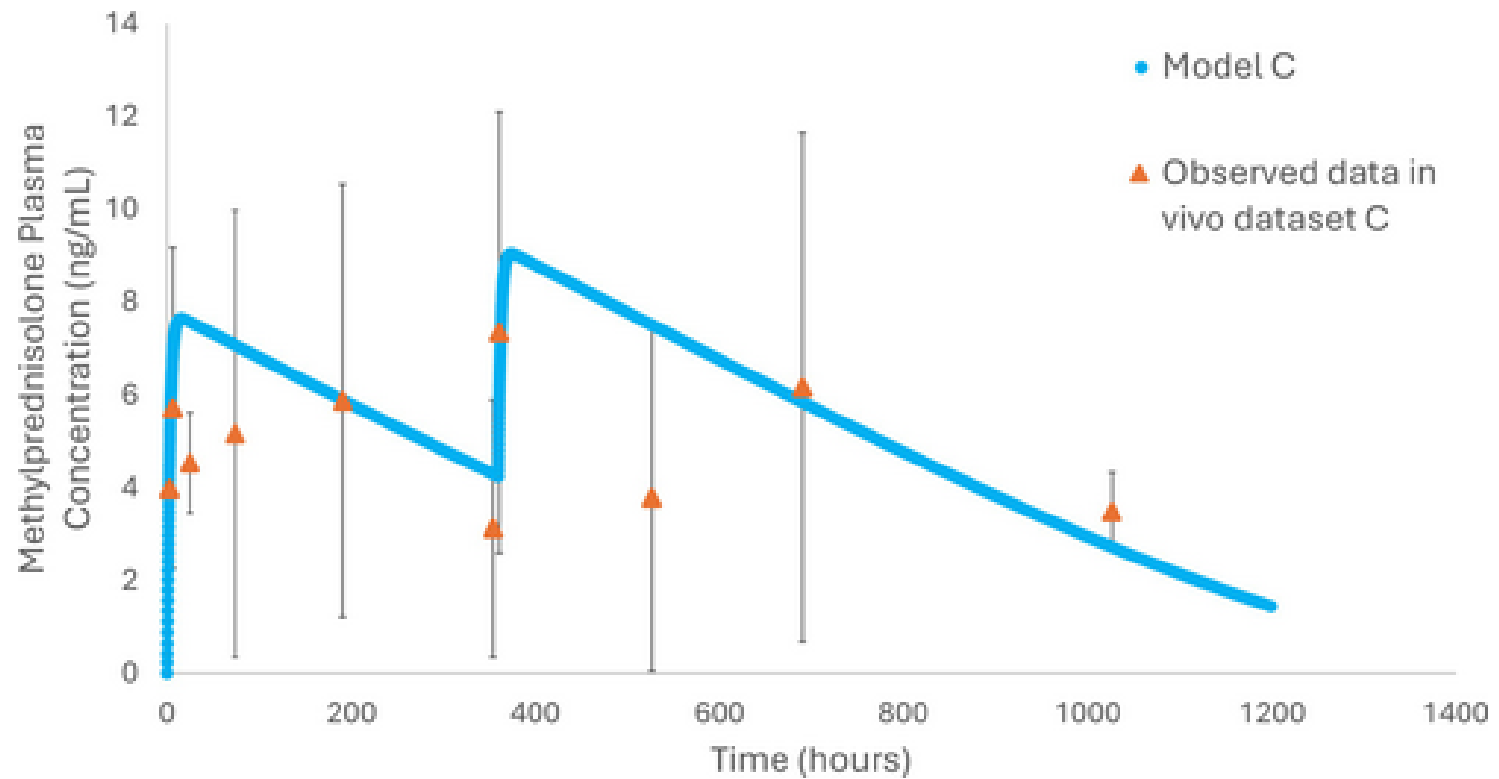


Figure 5: External validation of simulated plasma methylprednisolone (ng/mL) versus time (hours) of model C (blue) and in vivo dataset C (orange) 30. This dataset assess